EduFin SQL Business Challenge

Phase 1 — Monday Submission

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STEP 1A — Understand the Data Structure

```
SELECT
    loan_status AS `Loan Status`,
    COUNT(*) AS `Loan Count`,
    CONCAT(ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1), '%') AS `Percentage`
FROM loans
GROUP BY loan_status
ORDER BY `Loan Count` DESC
```

Loan Status	Loan Count	Percentage
Active	3965	79.3%
Defaulted	590	11.8%
Overdue	273	5.5%
Closed	172	3.4%

STEP 1B — Basic Portfolio Metrics

```
SELECT
    COUNT(DISTINCT customer_id) AS `Total Unique Customers`,
    COUNT(*) AS `Total Loans`,
    CONCAT(ROUND(SUM(loan_amount) / 10000000.0, 2), ' Cr') AS `Total Portfolio Value`,
    CONCAT(ROUND(AVG(loan_amount) / 100000.0, 2), ' L') AS `Average Loan Amount`
FROM loans;
```

Total Unique Customers	Total Loans	Total Portfolio Value	Average Loan Amount
3000	5000	204.82 Cr	4.1 L

STEP 1C — Calculate Risk Metrics

```
SELECT
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS `Defaulted Loans`,
    COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) AS `Overdue Loans`,
```

```
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Default Rate %`,
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN 1 END) *
100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Portfolio at Risk %`
FROM loans;
```

Defaulted Loans	Overdue Loans	Default Rate %	Portfolio at Risk %
590	273	11.80%	17.26%

STEP 1D — Financial Impact (Crores Format)

```
SELECT
    loan_status AS `Loan Status`,
    CONCAT(ROUND(SUM(loan_amount) / 10000000.0, 2), ' Cr') AS `Loan Value (Cr)`
FROM loans
GROUP BY loan_status
HAVING loan_status IN ('Overdue', 'Defaulted');

SELECT
    CONCAT(
        ROUND(
            SUM(CASE WHEN loan_status IN ('Overdue', 'Defaulted') THEN loan_amount
END)
            / 10000000.0, 2),
        ' Cr')
    AS `Total At Risk Value (Defaulted + Overdue)`
FROM loans;
```

Loan Status	Loan Value (Cr)
Defaulted	24.24 Cr
Overdue	10.94 Cr

Total At Risk Value (Defaulted + Overdue)
35.18 Cr

STEP 1E — Executive Dashboard (Final Consolidated Query)

```
SELECT
    COUNT(*) AS `Total Loans`,
    -- For Active Loans
    COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) AS `Active Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Active Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Active' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Active Portfolio Value`,
    -- For Closed Loans
    COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) AS `Closed Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Closed Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Closed' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Closed Portfolio Value`,
    -- For Defaulted Loans
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS `Defaulted Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Defaulted Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Defaulted Portfolio Value`,
    -- For Overdue Loans
    COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) AS `Overdue Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Overdue Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Overdue' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Overdue Portfolio Value`,
    -- Other Metrics
    COUNT(DISTINCT customer_id) AS `Total Customers`,
    CONCAT(ROUND(SUM(loan_amount) / 10000000.0, 2), ' Cr') AS `Total Portfolio Value`,
    CONCAT(ROUND(AVG(loan_amount) / 100000.0, 2), ' L') AS `Average Loan Amount`,
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN 1 END) *
100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Portfolio at Risk %`,
    CONCAT(
        ROUND(
            SUM(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN loan_amount
END) / 10000000.0,
            2
        ),
        ' Cr'
    ) AS `Total at Risk Value`,
    CASE
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 5.00 THEN 'HEALTHY'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 10.00 THEN 'MODERATE RISK'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 15.00 THEN 'HIGH RISK'
        ELSE 'CRITICAL'
    END AS `Portfolio Health Status`,
    CASE
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 12.00
            THEN 'Crisis team + external recovery agencies'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 10.00
            THEN 'Enhanced collection + review underwriting'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 5.00
            THEN 'Enhanced monitoring required'
        ELSE 'Standard monitoring protocols'
    END AS `Recommend Board Action`
FROM loans;
```

```
+----------------+--------------+--------------------+------------------+------------------+------------------+----------------------+----------------------+
| Portfolio Value|Total Customers|Total Portfolio Value|Average Loan Amount|Portfolio at Risk %|Total at Risk Value|Portfolio Health Status|Recommend Board Action|
+----------------+--------------+--------------------+------------------+------------------+------------------+----------------------+----------------------+
        10.94 Cr|          3000|           204.82 Cr|             4.1 L|            17.26%|          35.18 Cr|              HIGH RISK|  Enhanced collecti...|
+----------------+--------------+--------------------+------------------+------------------+------------------+----------------------+----------------------+

The portfolio has 35.18 Cr (17.26%) at risk from 590 defaulted and 273 overdue loans, indicating HIGH RISK. However, 162.41 Cr
(79.3%) remains active and performing well. Immediate action is required to recover overdue amounts and prevent further defaults.
```

Custom Column

```
CASE
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
5.00 THEN 'Stable'
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
8.00 THEN 'Monitor'
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
12.00 THEN 'Alert'
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
16.00 THEN 'Urgent'
    ELSE 'Emergency'
END AS risk_tier
```

> I designed EduFin's custom threshold to count both defaulted (590) and overdue (273) loans together as "effective default rate" because overdue loans are defaults-waiting-to-happen—they will become defaults within 6-12 months. This changes the picture from 11.80% (looks manageable) to 17.26% (clearly TIER 4: Crisis), forcing honest board-level action instead of complacency. Standard thresholds hide this future risk by only counting existing defaults; my threshold makes it visible today.

WHY Gate

1. Does your "Financial Impact" (Step 1D) include only defaulted loans, or does it include "Overdue" as well? Why did you make that choice?

> My financial impact (step 1D) includes both defaulted and overdue loans. It shows that the company has 35.18 Crores (Defaulted + Overdue) in total at-risk capital, broken down as 24.24 Crores in Defaulted loans and 10.94 Crores in Overdue loans. Both statuses are included because they represent capital the company may not recover, making them total value at risk financially.

2. If the CEO asks if this crisis is "temporary," what specific data point from your analysis would you point to as a warning sign that it's systemic?

> I would point out the 17.26% Portfolio at Risk (PAR) as a systemic warning signal because it is not just a snapshot of missed payments, it reflects a structural breakdown in repayment health across the portfolio, not isolated borrower behavior.

3. What is the single most important number in this workbook that the board should see first? Why that one?

> The single most importnat number in this workbook is "11.80% default rate", as it is the factor that indicates our entire situation as HIGH RISK against the industry benchmarks. Also, the CEO has explicitly asked us about the default rate (What's our overall default rate?) in the BRD.

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